HARD · GEOMETRY AND TRIGONOMETRY

# Right triangles and trigonometry

01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

**Q1** GEOMETRY AND TRIGONOMETRY

$$RS = 440$$

$$ST = 384$$

$$TR = 584$$

The side lengths of right triangle $RST$ are given. Triangle $RST$ is similar to triangle $UVW$, where $S$ corresponds to $V$ and $T$ corresponds to $W$. What is the value of $\tan W$?

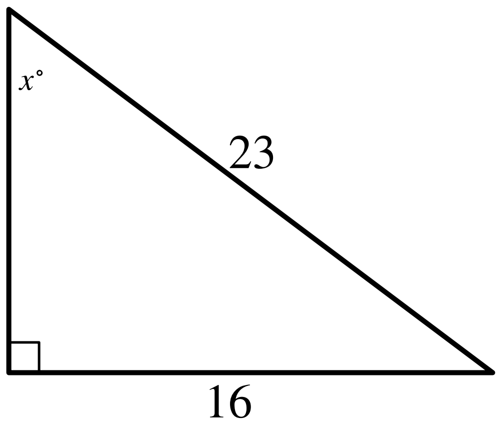
A. $\frac{48}{73}$

B. $\frac{55}{73}$

C. $\frac{48}{55}$

D. $\frac{55}{48}$

**Q2** GRID-IN GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x°.
- The length of the hypotenuse is 23.
- The length of the leg opposite the angle with measure x° is 16.
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\sin x°$?

STUDENT-PRODUCED RESPONSE

| | | | |
|---|---|---|---|
| . | / . | / . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GEOMETRY AND TRIGONOMETRY

For two acute angles, $\angle Q$ and $\angle R$, $\cos Q = \sin R$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 61$ and $4x + 4$, respectively. What is the value of $x$?

A. 5

B. 19

C. 23

D. 29

Q4

GEOMETRY AND TRIGONOMETRY

Which of the following expressions is equivalent to $\sin 24° \cos 66° + \cos 24° \sin 66°$?

A. $2\cos 66° \sin 24°$

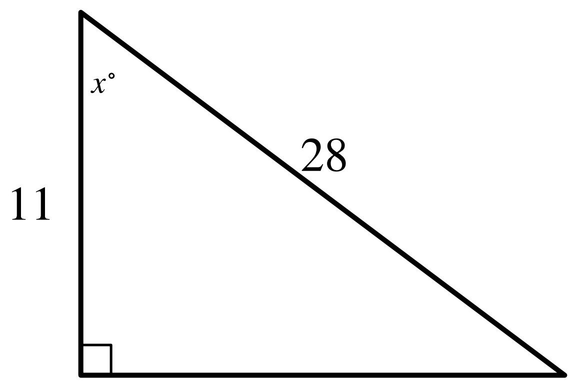
B. $2\cos 66° + 2\cos 24°$

C. $\cos 66°^2 + \cos 24°^2$

D. $\cos 66°^2 + \sin 24°^2$

Q5

GRID-IN GEOMETRY AND TRIGONOMETRY

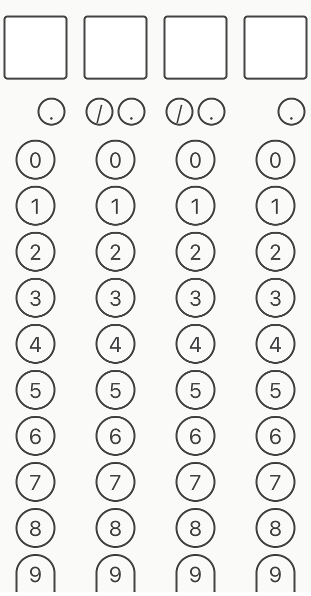



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x°.
- The length of the hypotenuse is 28.
- The length of the leg adjacent to the angle with measure x° is 11.
- A note indicates the figure is not drawn to scale.

In the triangle shown, what is the value of $\cos x°$?

STUDENT-PRODUCED RESPONSE



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

GRID-IN GEOMETRY AND TRIGONOMETRY

A triangle with angle measures 30°, 60°, and 90° has a perimeter of $18+6\sqrt{3}$. What is the length of the longest side of the triangle?

STUDENT-PRODUCED RESPONSE

| | | | |
|---|---|---|---|
| . | / . | / . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

GRID-IN GEOMETRY AND TRIGONOMETRY

Triangle $ABC$ is similar to triangle $DEF$, where angle $A$ corresponds to angle $D$ and angle $C$ corresponds to angle $F$. Angles $C$ and $F$ are right angles. If $\tan A = \frac{50}{7}$, what is the value of $\tan E$?

STUDENT-PRODUCED RESPONSE

| | | | |
|---|---|---|---|
| . | / . | / . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN GEOMETRY AND TRIGONOMETRY

A rectangle is inscribed in a circle, such that each vertex of the rectangle lies on the circumference of the circle. The diagonal of the rectangle is twice the length of the shortest side of the rectangle. The area of the rectangle is $1{,}089\sqrt{3}$ square units. What is the length, in units, of the diameter of the circle?

STUDENT-PRODUCED RESPONSE

| | | | |
|---|---|---|---|
| . | / . | / . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GEOMETRY AND TRIGONOMETRY

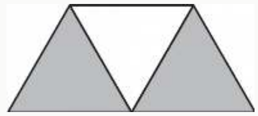
In triangle $ABC$, angle $B$ is a right angle. The length of side $AB$ is $10\sqrt{37}$ and the length of side $BC$ is $24\sqrt{37}$. What is the length of side $AC$?

A. $14\sqrt{37}$

B. $26\sqrt{37}$

C. $34\sqrt{37}$

D. $\sqrt{34 \cdot 37}$



A graphic designer is creating a logo for a company. The logo is shown in the figure above. The logo is in the shape of a trapezoid and consists of three congruent equilateral triangles. If the perimeter of the logo is 20 centimeters, what is the combined area of the shaded regions, in square centimeters, of the logo?

A. $2\sqrt{3}$

B. $4\sqrt{3}$

C. $8\sqrt{3}$

D. 16